

ppo (plain)

trained, not reported separately

stored weights byte-identical,
so the implementation term is
0 by verification, not assumption

ctrl

reported as PPO (baseline)
cost critic present
 λ ceiling = 0

activate
the budget

cppo

cost critic present
 λ free
budget $d = 25$

tighten
the budget

cppo15

cost critic present
 λ free
budget $d = 15$

constraint term
(the effect being claimed)

budget sensitivity
(does the effect scale with d ?)

Five seeds per arm (1, 3, 4, 52, 54), so fifteen trained policies. Every quantity is reported with its spread across seeds.